Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 1 of 20 · find part d in this mark scheme extract

3 a	$8\left(x^2+\frac{10}{8}x\right)-3$	M1
	$8\left(x+\frac{5}{8}\right)^2-\frac{200}{64}-3=8\left(x+\frac{5}{8}\right)^2-\frac{49}{8}$	M1
	So $A=8$ $B=\frac{5}{8}$ $C=-\frac{49}{8}$	A1 (3)
b (i)	$x=-\frac{5}{8}$	B1 ft
(ii)	$f(x)_{min}=-\frac{49}{8}$	B1 ft (2)
c	$8\left(x+\frac{5}{8}\right)^2-\frac{49}{8}=0\Rightarrow x=-\frac{3}{2}\quad x=\frac{1}{4}$	M1 A1 (2)
d	$8x^2+10x-3=2x+13\Rightarrow 8x^2+8x-16=0$ oe	M1
	$x^2+x-2=0=(x+2)(x-1)\Rightarrow x=-2,\ x=1$	M1 A1
	$y=9$ $y=15$ Coordinates are $(-2,9)$ and $(1,15)$	A1 (4)
e	Correct curve or line Correct curve and line	B1 B1 (2)
		Total 13 marks

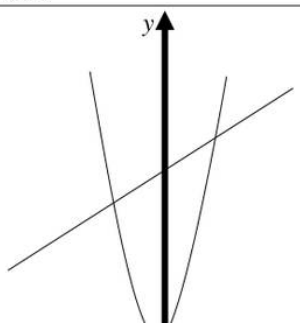
Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 2 of 20 · find part d in this mark scheme extract

(b) (i)	B1ft	For $x=-\frac{5}{8}$ ft their $-B$ Allow differentiation: $\frac{dy}{dx}=16x+10=0\Rightarrow x=-\frac{10}{16}$ which must be correct.
(ii)	B1ft	For $f(x)_{min}=-\frac{49}{8}$ ft their C If they differentiate, allow a ft from their differentiation.
	Incorrect or no labelling of parts. If the responses are labelled incorrectly, mark as labelled. If there is no labelling, treat the first as (i) and the second as (ii) and mark accordingly.	
(c)	M1	For setting the given f(x) = 0 and solving using any method. See General Guidance.
	A1	For $x=-\frac{3}{2}$ and $x=\frac{1}{4}$
(d)	M1	For setting $8x^2+10x-3=2x+13$ and forming a 3TQ which as a minimum must be $8x^2+8x\pm C$ where C is a constant, or any simplification, e.g., $x^2+x\pm K$ where K is a constant.
	M1	For attempting to solve their 3TQ. See General Guidance.
	A1	For **both** $x=-2$, $x=1$
	A1	For $(-2,9)$ and $(1,15)$ Accept $x=-2$, $y=9$ and $x=1$, $y=15$ paired correctly.
	ALT – solves the equation in y	
	M1	Substitutes $x=\frac{y-13}{2}$ into $y=8x^2+10x-3$ and forms a 3TQ which as a minimum must be of the form $2y^2-36y\pm X \quad X\neq 0$ $\Rightarrow y=8\left(\frac{y-13}{2}\right)^2+10\left(\frac{y-13}{2}\right)-3\Rightarrow 2y^2-36y+75=0$
	M1	For attempting to solve their 3TQ. See General Guidance.
	A1	For **both** $y=9$ and $y=15$
	A1	For $(-2,9)$ and $(1,15)$ Accept $x=-2$, $y=9$ and $x=1$, $y=15$ paired correctly.
(e)	B1	For correct curve (intersections with x-axis are at $(-1.5, 0)$ and $(0.25, 0)$) **or** line (Intersections are $(-6.5, 0)$ and $(0, 13)$ drawn. **We do not need to see any of these points marked. These are for guidance only.** The question asks for a sketch, accept a reasonable attempt. • Accept a positive quadratic curve with the minimum point below the x-axis, and one branch either side of the y-axis. • Accept a straight line with a positive gradient where the intersection with the y-axis is positive.
	B1	For correct curve **and** line drawn

SKETCH OF CURVE AND LINE



Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 3 of 20 · find part d in this mark scheme extract

Question number	Scheme	Marks
4	When $x=\frac{\pi}{2}$ $y=\frac{\pi^3}{8}$ So $\left(\frac{\pi}{2},\frac{\pi^3}{8}\right)$	B1
	$\frac{dy}{dx}=3x^2\sin x+x^3\cos x$	M1 A1 A1
	When $x=\frac{\pi}{2}$ $\frac{dy}{dx}=3\left(\frac{\pi}{2}\right)^2\sin\left(\frac{\pi}{2}\right)+\left(\frac{\pi}{2}\right)^3\cos\left(\frac{\pi}{2}\right)=\left[,\frac{3\pi^2}{4},\right]$	M1
	$y-\frac{\pi^3}{8}=\frac{3\pi^2}{4}\left(x-\frac{\pi}{2}\right)$	M1
	$y=\frac{3}{4}\pi^2x-\frac{1}{4}\pi^3$	A1
		Total 7 marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 4 of 20 · find part d in this mark scheme extract

Mark	Notes
Note: In this question, all substitution of angle values must be in Radians only.	
B1	For obtaining $y = \frac{\pi^3}{8}$ [allow $\left(\frac{\pi}{2}\right)^3$ and also awrt $y = 3.88$]
M1	For an attempt to use the product rule. The definition of an attempt is as follows: • There must be a correct attempt to differentiate both terms. $\sin x \Rightarrow \cos x \quad x^3 \Rightarrow ax^2$ where $x \neq 0$ • The correct formula must be used. i.e., it must be a sum of their two terms.
A1	At least one term must be correct. Either $3x^2 \sin x$ or $x^3 \cos x$
A1	For $3x^2 \sin x + x^3 \cos x$ oe Ignore any subsequent simplification once you have seen the correct answer – even if the simplification is incorrect.
M1	For substitution of $\frac{\pi}{2}$ into their $\frac{dy}{dx} = \left['\frac{3}{4}\pi^2'\right]$ provided it is a changed expression. Allow a value of awrt 7.4(0) **NOTE:** You must see a full substitution of $\frac{\pi}{2}$ into their $\frac{dy}{dx}$ if their expression for $\frac{dy}{dx}$ is incorrect.
M1	For a correct method for finding the equation of a line using their value of y, dy/dx and the given x [allow $x = 1.57\ldots$]. This must be applied correctly. Either uses the formula with their values, or if uses $y = mx + c$ they must reach a value for c before this mark can be awarded. Do **not** allow processing errors to find the value of c
A1	For $y = \frac{3}{4}\pi^2 x - \frac{1}{4}\pi^3$ Allow $y = 7.4(0)x - 7.75$ or better values. Do not allow a mixture of decimals and exact values.

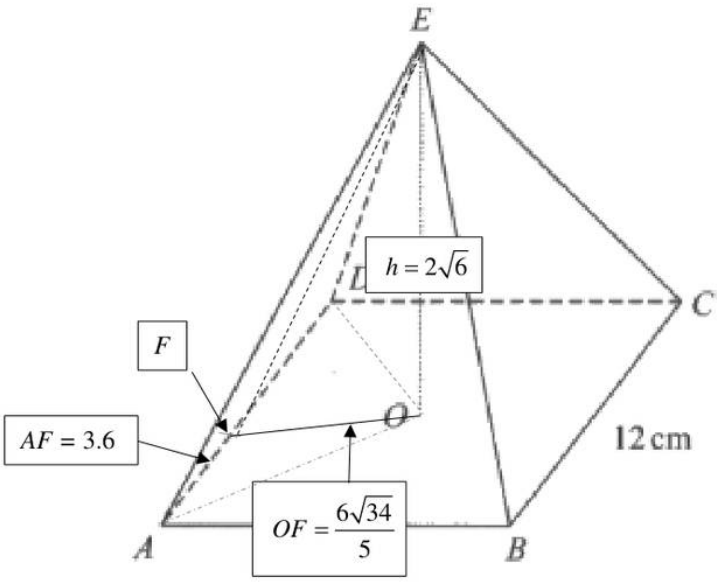
Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 5 of 20 · find part d in this mark scheme extract

Question number	Scheme	Marks
5 a	$OC = \frac{1}{2}\sqrt{12^2+12^2} = \left[6\sqrt{2}\right]$ or $AC = \sqrt{12^2+12^2} = \left[12\sqrt{2}\right]$	M1
	$h = EO = 6\sqrt{2} \times \frac{\sqrt{3}}{3} = 2\sqrt{6}$ oe	M1 A1
		(3)
b	Midpoint of AD to $F = \left(6 - \frac{12}{1+4}\right)$ [= 3.6]	M1
	$OF = \sqrt{6^2+3.6^2} = \frac{6\sqrt{34}}{5}$	M1
	$\tan\theta = \frac{2\sqrt{6}}{\frac{6\sqrt{34}}{5}} = 35^\circ$	M1 A1
		(4)
		Total 7 marks

USEFUL SKETCH



Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 6 of 20 · find part d in this mark scheme extract

Part	Mark	Notes
(a)	M1	For using a correct Pythagoras theorem to find either OC, OA $\left(6\sqrt{2}\right)$ or AC $\left(12\sqrt{2}\right)$
	M1	For using the correct trigonometry (tan ratio): $h = '6\sqrt{2}'\tan 30^\circ = \left[2\sqrt{6}\right]$ or equivalent. Eg $\tan 30 = \dfrac{h}{6\sqrt{2}} \Rightarrow h = ...$
	A1	For $2\sqrt{6}$ oe
(b)	M1	For correct expression for the midpoint of AD to F $\left(6 - \dfrac{12}{1+4}\right)$ or 3.6 oe seen
	M1	For the correct use of Pythagoras' to find OF $\dfrac{6\sqrt{34}}{5}$ oe [6.997..] ft their 3.6
	ALT	
	M1	Finds the length OF using cosine rule. $OF = \sqrt{2.4^2 + \left(6\sqrt{2}\right)^2 - 2\times 2.4\times 6\sqrt{2}\times\cos 45} = \left[\dfrac{6\sqrt{34}}{5}\right]$ OR $OF = \sqrt{9.6^2 + \left(6\sqrt{2}\right)^2 - 2\times 9.6\times 6\sqrt{2}\times\cos 45} = \left[\dfrac{6\sqrt{34}}{5}\right]$ OR $\cos 45^\circ = \dfrac{9.6^2 + \left(6\sqrt{2}\right)^2 - OF^2}{2\times 9.6\times 6\sqrt{2}} \Rightarrow OF = ...$ oe [6.997..]
	M1	For the correct evaluation of their cosine rule. $\left(\dfrac{6\sqrt{34}}{5}\text{ oe}\right)$ Accept awrt 7.00 [6.997142…]
	dM1	For the correct use of $\tan\theta = \dfrac{'EO'}{'OF'}$ This mark is dependent on both previous M marks. They must have a valid method to find OF for the award of this mark.
	A1	For 35(°) or better (Calculator value is 34.997…°.)
	Note: There are other methods – if unsure, send to Review.	

Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 7 of 20 · find part d in this mark scheme extract

Question number	Scheme	Marks
6 a	$r=\frac{q(2p-3)}{q(2p+3)}=\frac{q(2p+3)}{q(4p+1)}$	M1
	$(4p+1)(2p-3)=(2p+3)^2$	dM1
	$2p^2-11p-6=0$	A1
	$(2p+1)(p-6)=0$	M1
	$p=-\frac{1}{2}$ or $p=6$	A1 (5)
b	When $p=6$ $r=\frac{3}{5}$ and $U_1=q\left(4\times'6'+1\right)=\left(25q\right)$	M1
	$S_\infty=\frac{25q}{\frac{2}{5}}=250\Rightarrow q=4$	dM1 A1 (3)
		Total 8 marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 8 of 20 · find part d in this mark scheme extract

Part	Mark	Notes
(a)	M1	For $r = \frac{\text{3rd term}}{\text{2nd term}} = \frac{\text{2nd term}}{\text{1st term}}$
	dM1	For attempting to remove the denominators/simplifying the expression to attempt to obtain a 3TQ. Allow one processing error. Allow a minimally acceptable $4p^2 \pm Xp \pm Y = (0)$ or $2p^2 \pm Pp \pm Q = (0)$ $X, Y, P, Q \neq 0$ **Note:** This mark is dependent on the previous M mark.
	A1	For obtaining $2p^2 - 11p - 6 = (0)$ or equivalent. Their working will give them $4p^2 - 22p - 12 = (0)$ before further simplification
	M1	For a valid attempt to solve the 3TQ. Note this is an independent mark for their 3TQ. See General Guidance. The 3TQ must have come from some attempted manipulation involving $q(2p-3)$, $q(2p+3)$ and $q(4p+1)$
	A1	For $p = -\frac{1}{2}$ or $p = 6$
(b)	M1	For finding: • A value of r for their p. Allow any value of r $r \neq 0$ E.g. $r = \frac{(2\times'6'-3)}{(2\times'6'+3)} = ...$ $\left(\text{Note: for } p = -\frac{1}{2} \quad r = -2\right)$ • A value for U_1 $U_1 = q(4\times'6'+1) = ...$ $\left(\text{Note: for } p = -\frac{1}{2}, U_1 = -q\right)$ You may see a = 100 after later working to find the sum to infinity. If their values of p are incorrect, we **must** see working here. **NOTE:** This is a B mark in Epen.
	dM1	For use of $S_\infty = \frac{a}{1-r} = 250$ with their r and their U_1 provided $\lvert r\rvert < 1$ If they use the formula for the sum to infinity on an $\lvert r\rvert > 1$ withhold this mark even if they use it on one valid and one invalid attempt. **Note:** This mark is dependent on the previous M mark.
	A1	For $q = 4$

Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 9 of 20 · find part d in this mark scheme extract

Question number	Scheme	Marks
7 a	$\frac{dy}{dx} = 2e^{2x}\cos 2x - 2e^{2x}\sin 2x$	M1 A1 A1
	$\frac{dy}{dx} = 2y - 2e^{2x}\sin 2x$ *	A1 cso (4)
b	$\frac{d^2y}{dx^2} = 2\frac{dy}{dx} - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$	M1 A1
	$\frac{d^2y}{dx^2} = 2\frac{dy}{dx} + 2\left(\frac{dy}{dx} - 2y\right) - 4y$	dM1ddM1
	$\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$ *	A1 cso (5)
		Total 9 marks

Part	Mark	Notes
(a)	M1	For an attempt to use the product rule. The definition of an attempt is as follows: • There must be a correct attempt to differentiate both terms. $\cos 2x \Rightarrow \pm 2\sin 2x \quad e^{2x} \Rightarrow 2e^{2x}$ where $x \neq 0$ • The terms must be added The correct formula must be used.
	A1	For one term correct
	A1	For both terms correct
	A1 cso	For obtaining the given result with no errors seen.
(b)	**Method 1**	
	M1	For attempting to differentiate $\frac{dy}{dx}$ to obtain $\frac{d^2y}{dx^2}$ $\frac{d^2y}{dx^2} = 4e^{2x}\cos 2x - 4e^{2x}\sin 2x - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$ For this mark accept either: $\left(4e^{2x}\cos 2x - 4e^{2x}\sin 2x\right)$ **OR** $-4e^{2x}\sin 2x - 4e^{2x}\cos 2x$
	A1	For the correct $\frac{d^2y}{dx^2}$ (This is an M mark in Epen)
	dM1	For simplifying $\frac{d^2y}{dx^2}$ to obtain $\frac{d^2y}{dx^2} = -8e^{2x}\sin 2x$ (A mark in Epen) **This mark is dependent on the first M mark**

Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 10 of 20 · find part d in this mark scheme extract

	ddM1	For using the substitution $-2e^{2x}\sin 2x = \frac{dy}{dx} - 2y \Rightarrow -8e^{2x}\sin 2x = 4\frac{dy}{dx} - 8y$ **This mark is dependent on BOTH previous M marks.**
	A1cso	For obtaining the given result with no errors seen. $\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$
	Method 2	
	M1	For attempting to differentiate $\frac{dy}{dx}$ to obtain $\frac{d^2y}{dx^2}$ $\frac{d^2y}{dx^2} = 2\frac{dy}{dx} - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$ For this mark accept either: $2\frac{dy}{dx}$ or $-4e^{2x}\sin 2x - 4e^{2x}\cos 2x$
	A1	For the correct $\frac{d^2y}{dx^2}$ **This is an M mark in Epen**
	dM1	For substituting $-4e^{2x}\cos 2x \Rightarrow -2y$ **This is an A mark in Epen**
	ddM1	For using the substitution $-4e^{2x}\sin 2x = 2\frac{dy}{dx} - 4y$
	A1	For obtaining the given result with no errors seen. $\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$
	Method 3 – Works form LHS and RHS together.	
	M1	For attempting to differentiate $\frac{dy}{dx}$ to obtain $\frac{d^2y}{dx^2}$ $\frac{d^2y}{dx^2} = 4e^{2x}\cos 2x - 4e^{2x}\sin 2x - 4e^{2x}\sin 2x - 4e^{2x}\cos 2x$ For this mark accept either: $\left(4e^{2x}\cos 2x - 4e^{2x}\sin 2x\right)$ **OR** $-4e^{2x}\sin 2x - 4e^{2x}\cos 2x$
	A1	For the correct $\frac{d^2y}{dx^2}$ **This is an M mark in Epen**
	dM1	For simplifying $\frac{d^2y}{dx^2}$ to obtain $\frac{d^2y}{dx^2} = -8e^{2x}\sin 2x$ (A mark in Epen) **This mark is dependent on the first M mark**
	ddM1	Multiplies out the given expression in (b) $\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y = 4\left(2e^{2x}\cos 2x - 2e^{2x}\sin 2x\right) - 8\left(e^{2x}\cos 2x\right)$ $= 8e^{2x}\cos 2x - 8e^{2x}\sin 2x - 8e^{2x}\cos 2x$ $= -8e^{2x}\sin 2x$

Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 11 of 20 · find part d in this mark scheme extract

		This mark is dependent on BOTH previous M marks.
	A1	For a conclusion. A simple # sign, 'shown', 'QED' or underlining is sufficient.

Question number	Scheme	Marks
8	$\alpha+\beta=4k\sqrt{2}$ and $\alpha\beta=2k^4-1$	B1 B1
	$(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta\Rightarrow\left(4k\sqrt{2}\right)^2=66+2\left(2k^4-1\right)$	M1 A1
	$k^4-8k^2+16=0$	M1
	$\left(k^2-4\right)^2=0\Rightarrow k=2$	M1 A1
	$(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3$ **or** $\alpha^3+\beta^3=(\alpha+\beta)\left(\alpha^2+\beta^2-\alpha\beta\right)$	M1
	$\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ **or** $\alpha^3+\beta^3=(\alpha+\beta)(66-\alpha\beta)$	A1
	$\alpha^3+\beta^3=\left(8\sqrt{2}\right)^3-3\times31\times8\sqrt{2}=280\sqrt{2}$ **or** $8\sqrt{2}(66-31)=280\sqrt{2}$	M1
	p=280	A1 (11)
		Total 11 marks

Mark	Notes
B1	For **either** $\alpha+\beta=4k\sqrt{2}$ **or** $\alpha\beta=2k^4-1$
B1	For **both** $\alpha+\beta=4k\sqrt{2}$ **and** $\alpha\beta=2k^4-1$
M1	For the correct algebra on $\alpha^2+\beta^2$ (in any order) **and** substitution of their values of $\alpha\beta$ and $\alpha+\beta$ providing both sum and product are in terms of k. $(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta\Rightarrow\alpha^2+\beta^2=\left('4k\sqrt{2}'\right)^2-2\left('2k^4-1'\right)$
A1	For obtaining $\left(4k\sqrt{2}\right)^2=66+2\left(2k^4-1\right)$ in any order.
M1	For simplifying to form a 3TQ in k^4 i.e., $4k^4-32k^2+64=0$ oe Accept as a minimum $4k^4-32k^2\pm Q=(0)\quad Q\neq0$
M1	For factorising or solving the 3TQ using any valid method. See General Guidance.
A1	For $k=2$ If they also give $k=-2$ withhold this mark.

Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 12 of 20 · find part d in this mark scheme extract

Method 1	
M1	For expanding $(\alpha+\beta)^3=\alpha^3+3\alpha^2\beta+3\alpha\beta^2+\beta^3$ or $\alpha^3+\beta^3=(\alpha+\beta)(\alpha^2+\beta^2-\alpha\beta)$
A1	For obtaining $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$ or $\alpha^3+\beta^3=(\alpha+\beta)(66-\alpha\beta)$ This must be such that $\alpha\beta$ and $\alpha+\beta$ can be substituted in directly.
M1	For substitution of $\alpha+\beta$ and $\alpha\beta$ for their positive value of k into a **correct** expansion of $\alpha^3+\beta^3$ **NOTE:** If they do not obtain $k=2$, then full substitution of their numerical value for k into $\alpha+\beta$ **and** $\alpha\beta$ must be seen for the award of this mark. For example: $\alpha^3+\beta^3=\left(4\times\text{'their } k\text{'}\sqrt{2}\right)^3-3\times\left(2\times\text{'their } k^4\text{'}-1\right)\times\left(4\times\text{'their } k\text{'}\sqrt{2}\right)$
A1	For $p=280$
Method 2	
M1	Finds the exact value of α and β Solves the equations $\left(\alpha\beta=2k^4-1 \text{ and } \alpha+\beta=4k\sqrt{2}\right)$ simultaneously to give a value for α and β $\alpha=\frac{31}{\beta}\Rightarrow\alpha+\beta=\frac{31}{\beta}+\beta=8\sqrt{2}\Rightarrow\beta^2-8\sqrt{2}\beta+31=0$ $\Rightarrow\beta=\ldots\quad\alpha=\ldots$
A1	For $\alpha=1+4\sqrt{2}\quad\beta=-1+4\sqrt{2}$ OR $\beta=1+4\sqrt{2}\quad\alpha=-1+4\sqrt{2}$
M1	Substitutes these values into $\alpha^3+\beta^3=p\sqrt{2}$ to find a value for p $\left(1+4\sqrt{2}\right)^3+\left(-1+4\sqrt{2}\right)^3=24\sqrt{2}+256\sqrt{2}=280\sqrt{2}\Rightarrow p=\ldots$
A1	$p=280$

Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 13 of 20 · find part d in this mark scheme extract

Question number	Scheme	Marks
9		
	$\frac{dA}{dt} = 0.45$	B1
	$V = x^3 \Rightarrow \frac{dV}{dx} = 3x^2$	B1
	$A = 6x^2 \Rightarrow \frac{dA}{dx} = 12x$	B1
	$384 = 6x^2 \Rightarrow x = 8$	M1
	$\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dA} \times \frac{dA}{dt}$	M1
	$\frac{dV}{dt} = 3x^2 \times \frac{1}{12x} \times 0.45 \left[= \frac{9x}{80} \right]$ oe	dM1
	When $x = 8$ $\frac{dV}{dt} = 0.9\,\text{cm}^3/\text{s}$	A1
		Total 7 marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 14 of 20 · find part d in this mark scheme extract

Mark	Notes
B1	For $\frac{dA}{dt} = 0.45$ seen anywhere in their working Accept other letters, for example S for the area $\frac{dS}{dt} = 0.45$
B1	For $\frac{dV}{dx} = 3x^2$ Accept also other letters in place of x such as r for example.
B1	For $\frac{dA}{dx} = 12x$ Accept also other letters in place of x such as r for example.
M1	For setting $384 = 6x^2$ and proceeding to a correct method leading to a value of x Award this mark when they obtain $x^2 = 64 \Rightarrow x = ...$
M1	For a correct expression of the chain rule seen or implied. i.e., $\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dA} \times \frac{dA}{dt}$ They may complete this in two stages. So you may see for example: $\frac{dx}{dt} = \frac{1}{\frac{dA}{dx}} \times \frac{dA}{dt}$ **AND** $\frac{dV}{dt} = \frac{dx}{dt} \times \frac{dV}{dx}$
dM1	For substituting their values into a **correct** chain rule. $\frac{dV}{dt} = 3(8)^2 \times \frac{1}{12(8)} \times 0.45$ This mark is dependent on the previous M mark scored.
A1	For 0.9 (cm^3/s)

Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 15 of 20 · find part d in this mark scheme extract

Question number	Scheme	Marks
10 a (i)	$\sin 2\theta = \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta$ *	M1 A1 cso (2)
a (ii)	$\cos 2\theta = \cos^2\theta - \sin^2\theta = \cos^2\theta - (1-\cos^2\theta)$	M1 M1
	$= 2\cos^2\theta - 1$ *	A1 cso (3)
b	$\sin 2\theta - \tan\theta = 2\sin\theta\cos\theta - \dfrac{\sin\theta}{\cos\theta}$	M1
	$\sin\theta\left(2\cos\theta - \dfrac{1}{\cos\theta}\right)$	M1
	$\sin\theta\left(\dfrac{2\cos^2\theta - 1}{\cos\theta}\right) = \tan\theta\cos 2\theta$ *	dM1 A1cso (4)
c	$\tan x = 0$ so $x = 180$	B1
	$\cos 2x = 0$ so $2x = 90, 270, 450, 630$	M1
	$x = 45, 135, 225, 315$	A1 A1 (4)
		Total 13 marks

Part	Mark	Notes
(a)(i)	M1	For the correct use of: $\sin(A+B) = \sin A\cos B + \cos A\sin B \Rightarrow \sin 2\theta = \sin\theta\cos\theta + \sin\theta\cos\theta$ Allow any letter to be used for this mark.
	A1 Cso	For obtaining the given result $\sin 2\theta = 2\sin\theta\cos\theta$ or allow $\sin(\theta+\theta) = 2\sin\theta\cos\theta$
(a)(ii)	M1	For the correct use of: $\cos(A+B) = \cos A\cos B - \sin A\sin B$ $\Rightarrow \cos 2A = \cos^2 A - \sin^2 A$ or $\cos(A+A) = \cos A\cos A - \sin A\sin A$ Allow any letter to be used for this mark.
	M1	For use of $\sin^2 A + \cos^2 A = 1$ and substituting Allow any letter to be used for this mark.
	A1 cso	For obtaining the given result $\cos 2\theta = 2\cos^2\theta - 1$ or allow $\cos(\theta+\theta) = 2\cos^2\theta - 1$
(b)	M1	For correct use of $\sin 2\theta = 2\sin\theta\cos\theta$ **and** $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$

Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 16 of 20 · find part d in this mark scheme extract

	M1	For factorising the resulting expression to give $\sin\theta\left(2\cos\theta-\frac{1}{\cos\theta}\right)$ **and** using a common denominator of $\cos\theta$ correctly to obtain $\sin\theta\left(\frac{2\cos^2\theta-1}{\cos\theta}\right)$ oe e.g. $\frac{\sin\theta}{\cos\theta}\left(2\cos^2\theta-1\right)$
	dM1	For use of $2\cos^2\theta-1=\cos 2\theta$ **Note:** This mark is dependent on both previous M marks
	A1 cso	For obtaining the given result
	ALT - Allow candidates to work from both sides. For example.	
	M1	For correct use of $\sin 2\theta=2\sin\theta\cos\theta$ **and** $\tan\theta=\frac{\sin\theta}{\cos\theta}$ $\sin 2\theta-\tan\theta=\tan\theta\cos 2\theta$ $\Rightarrow 2\sin\theta\cos\theta-\frac{\sin\theta}{\cos\theta}=\frac{\sin\theta}{\cos\theta}\left(2\cos^2\theta-1\right)$
	M1	For using a common denominator on the LHS to give: $\frac{2\sin\theta\cos^2\theta-\sin\theta}{\cos\theta}=\frac{\sin\theta\left(2\cos^2\theta-1\right)}{\cos\theta}$
	dM1	Multiplies or factorises one side to make LHS = RHS For example; $\frac{2\sin\theta\left(\cos^2\theta-1\right)}{\cos\theta}=\frac{2\sin\theta\left(\cos^2\theta-1\right)}{\cos\theta}$
	A1 cso	Concludes RHS = LHS therefore shown. Note there must be no errors seen. If unsure, send to Review.
(c)	Some candidates are starting again without using the given expression in (b) and are getting their angles from cos x rather than cos $2x$ Allow the use of θ or any other variable throughout this part of the question. As a general principle of marking this part: • B1 – For only 180° seen. If 0° is included, then B0. • M1 – For either $2x=90,(270,450,630)$ **OR** $x=45^\circ$ or 135° or 225° or 315° or any other correct angle in or out of range. • A1 – For any two correct angles apart from 180° • A1 – For all four correct angles, apart form 180°	
	B1	For $x=180$ If $x=0$ included then B0
	M1	For obtaining $2x=90,(270,450,630)$
	A1	For any two correct angles from 45°, 135°, 225° or 315°
	A1	For all four angles from $\cos x$ or $\cos 2x$
	For extra angles within range, penalise the last A mark only. For example; If they have [coming from a correct method] $x=45°, 67.5°, 135°, 225°, 315°$ score M1A1A0 If they have $x=45°, 67.5°, 135°$ then award M1A1A0	

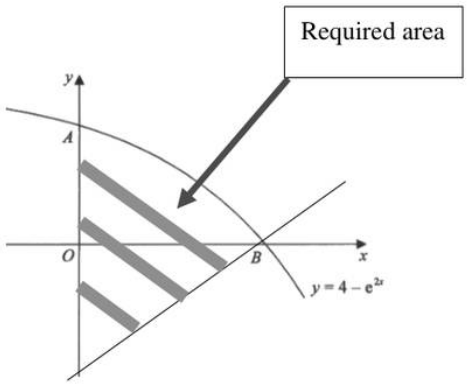
Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 17 of 20 · find part d in this mark scheme extract

Question number	Scheme	Marks
11 a	When $x = 0 \quad y = 3$	B1
	When $y = 0 \quad 0 = 4 - e^{2x} \Rightarrow \ln 4 = 2x \Rightarrow x = \ln 2$	M1 A1 (3)
b	$\frac{dy}{dx} = -2e^{2x}$	M1
	When $x = \ln 2 \quad \frac{dy}{dx} = -8$	M1
	$y - 0 = \frac{1}{8}(x - \ln 2) \Rightarrow y = \frac{1}{8}x - \frac{1}{8}\ln 2$	M1 A1 (4)
c	$\int_0^{\ln 2}\left(4 - e^{2x}\right)dx = \left[4x - \frac{1}{2}e^{2x}\right]_0^{\ln 2} = \left(4\ln 2 - \frac{1}{2}e^{2\ln 2}\right) - \left(0 - \frac{1}{2}e^{0}\right)$	M1M1M1
	$\frac{1}{2}(\ln 2)\left(\frac{1}{8}\ln 2\right) = \frac{1}{16}(\ln 2)^2$	M1M1
	$\frac{1}{8}\int_0^{\ln 2}\lvert x - \ln 2\rvert\,dx = \frac{1}{8}\left\lvert\frac{x^2}{2} - x\ln 2\right\rvert_0^{\ln 2} = \left\lvert\frac{(\ln 2)^2}{16} - \frac{(\ln 2)^2}{8}\right\rvert = \frac{(\ln 2)^2}{16}$	[M1M1]
	$4\ln 2 - \frac{3}{2} + \frac{1}{16}(\ln 2)^2 = 1.3$	M1 A1 (7)
		Total 14 marks

USEFUL SKETCH



Part	Mark	Notes

Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 18 of 20 · find part d in this mark scheme extract

(a)(i) (ii)	B1	For $y = 3$
	M1	For $\ln 4 = 2x$ or $\sqrt{4} = \sqrt{e^{2x}}$ seen explicitly
	A1	For $x = \ln 2$
(b)	M1	For differentiating the given expression. This must be correct for this mark. $\frac{dy}{dx} = -2e^{2x}$
	M1	For substituting ln2 into their $\frac{dy}{dx} = -2e^{2\ln 2} = [-8]$
	M1	For a correct method for finding the equation of a straight line using their numerical perpendicular gradient and $y = 0$ and $x = \ln 2$ Award when they substitute into a correct formula, or if they use $y = mx + c$ award when c is obtained. (accept a decimal value for c for this mark awrt $c = -0.087$)
	A1	For $y = \frac{1}{8}x - \frac{1}{8}\ln 2$ or $y = \frac{x}{8} - \frac{\ln 2}{8}$ in exact form only.
(c)	**Area under curve**	
	M1	For a correct statement for the area under the curve with correct limits. Accept the limits either way around. Ignore poor notation. This mark can be implied by later correct work. $\int_0^{\ln 2}\left(4 - e^{2x}\right)dx$
	M1	For a minimally acceptable attempt to integrate as follows. $\int_0^{\ln 2}\left(4 - e^{2x}\right)dx = \left[4x \pm \frac{e^{2x}}{2}\right]$
	M1	For substitution of both limits into their integral of the curve. $A = \left(4\ln 2 - \frac{1}{2}e^{2\ln 2}\right) - \left(0 - \frac{1}{2}e^{0}\right) = \left[4\ln 2 - \frac{3}{2}\right] = [1.27258....]$
	Area of the triangle	
	M1	**Method 1** For a statement of the area. $A = \frac{1}{2}(\ln 2)\left('\frac{1}{8}\ln 2'\right)$ **Method 2** For a statement of the area, with limits either way around. $A = \frac{1}{8}\int_0^{\ln 2}(x - \ln 2)\,dx$ ft their equation of the line.
	M1	For a correct method to evaluate the area of the triangle. **Method 1** $A = \frac{1}{2}(\ln 2)\left('\frac{1}{8}\ln 2'\right) = \frac{1}{16}(\ln 2)^2 = [0.03002.....]$ **Method 2** The integration and substitution must be correct for this mark $A = \frac{1}{8}\int_0^{\ln 2}\lvert x - \ln 2\rvert\,dx = \frac{1}{8}\left\lvert\frac{x^2}{2} - x\ln 2\right\rvert_0^{\ln 2} = \left\lvert\frac{(\ln 2)^2}{16} - \frac{(\ln 2)^2}{8}\right\rvert = \frac{(\ln 2)^2}{16}$

Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 19 of 20 · find part d in this mark scheme extract

	Combined area	
	M1	For the total final area using their values $4\ln 2-\frac{3}{2}+\frac{1}{16}(\ln 2)^2$ **This is an A mark in Epen.**
	A1	For awrt 1.3 Also accept the exact Area = $4\ln 2-\frac{3}{2}+\frac{1}{16}(\ln 2)^2$

Type MS · Where a line meets a curve (simultaneous equations)

Answer 1 of 19 · Question 3 part d · 4 marks

2023 JAN · Paper 02 · extract 20 of 20 · find part d in this mark scheme extract

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Type MS · Where a line meets a curve (simultaneous equations)

Answer 2 of 19 · Question 5 part whole · 6 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

5	$(x+2y=17)\quad x=\frac{36}{y}\quad \left(\text{or } y=\frac{36}{x}\right)$	M1
	$\frac{36}{y}+2y=17,\quad 36+2y^2=17y\quad (\text{or } 72+x^2=17x)$	M1
	$2y^2-17y+36\ (=0)\qquad (\text{or } x^2-17x+72=0)$	A1
	$(y-4)(2y-9)=0\qquad (\text{or } (x-8)(x-9))$	
	$y=4\quad x=9$	dM1A1
	$y=4\frac{1}{2}\quad x=8$	A1 (6)

M1	Rearrange $xy = 36$ to $x = ...$ or $y = ...$
M1	Eliminate x or y from the linear equation and obtain a 3TQ, $= 0$ not needed
A1	Correct 3TQ, terms in any order. $= 0$ not needed
M1	Solve their 3TQ by any valid method. Obtain at least one value for y or x
A1	**Either** 2 correct values for x or y **or** a correct (x, y) pair
A1	Both pairs correct and pairing clear.

ALT:	The following method may possibly be seen:	
	$xy + x + 2y = 53 \Rightarrow 36 + x + 2y = 53 \Rightarrow x + 2y = 17$ and $xy = 36$ or $x \times 2y = 72$	M1
	Hence x and $2y$ are the roots of the equation $z^2 - 17z + 72 = 0$	M1A1
	$(z - 9)(z - 8) = 0 \Rightarrow z = 9$ or 8	M1
	So $x = 8\quad y = 4.5$ or $x = 9\quad y = 4$	A1A1 [6]

M1	Substitute $xy = 36$ in the linear equation to obtain $x + 2y = 17$ and $xy = 36$ oe
M1	Obtain a 3TQ with roots x and $2y$
A1	Correct 3TQ
M1	Solve their 3TQ by any valid method. Obtain at least one value for *for the roots*
A1	**Either** 2 correct values for x or y **or** a correct (x, y) pair
A1	Both pairs correct and pairing clear.

	Special Case
	$x+2y=17\quad xy=36$ Use $xy=36$ in the other equation to obtain $x+2y=17$
	$\Rightarrow x=9\quad y=4$ By inspection:
	Score M1M0A0M1A1A0 (Must see $x+2y=17$; otherwise no marks)
	If the second answer is also obtained correctly by inspection, award all marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 2 of 19 · Question 5 part whole · 6 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract

Question Number	Scheme	Marks
	$\frac{dy}{}$	

Type MS · Where a line meets a curve (simultaneous equations)

Answer 3 of 19 · Question 2 part a · 4 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 4 · find part a in this mark scheme extract

2(a)	$6\times\left(-\frac{1}{2}\right)^2-\left(-\frac{1}{2}\right)q=5\Rightarrow\frac{3}{2}+\frac{q}{2}=5\Rightarrow q=7$	M1A1
	$y=px+9\Rightarrow 7=p\times-\frac{1}{2}+9\Rightarrow p=4$	M1A1 [4]
	ALT	
	$6x^2-x(px+9)=5\Rightarrow\frac{3}{2}-\frac{p}{4}+\frac{9}{2}-5=0\Rightarrow p=4$ $y=px+9\ \Rightarrow\ q=4\times-\frac{1}{2}+9\Rightarrow q=7$	[M1A1 M1A1]
	ALT 2	
	$6x^2-x(px+9)-5=0$ and $(2x+1)(ax+b)=0\Rightarrow 2ax^2+x(2b+a)+b=0$ $6-p=2a$ and $b=-5$ $-9=2b+a\ \Rightarrow\ -9=2\times-5+a\Rightarrow a=1$ $\therefore 6-p=2\Rightarrow p=4$ $q=4\times-\frac{1}{2}+9=7$	[M1A1 M1A1]
(b)	$6x^2-x(4x+9)=5\Rightarrow 6x^2-4x^2-9x-5=0\Rightarrow 2x^2-9x-5=0$	M1
	$2x^2-9x-5=0\Rightarrow(2x+1)(x-5)=0\Rightarrow x=5$	M1A1
	$y=4\times5+9=29\Rightarrow(5,\ 29)$	A1 [4]
		Total 8 marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 3 of 19 · Question 2 part a · 4 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks
2(a)	$6\times\left(-\frac{1}{2}\right)^2-\left(-\frac{1}{2}\right)q=5\Rightarrow\frac{3}{2}+\frac{q}{2}=5\Rightarrow q=7$	M1A1
	$y=px+9\Rightarrow 7=p\times-\frac{1}{2}+9\Rightarrow p=4$	M1A1 [4]
	ALT	
	$6x^2-x(px+9)=5\Rightarrow\frac{3}{2}-\frac{p}{4}+\frac{9}{2}-5=0\Rightarrow p=4$	[M1A1
	$y=px+9\ \Rightarrow q=4\times-\frac{1}{2}+9\Rightarrow q=7$	M1A1]
	ALT 2	
	$6x^2-x(px+9)-5=0$ and $(2x+1)(ax+b)=0\Rightarrow 2ax^2+x(2b+a)+b=0$ $6-p=2a$ and $b=-5$ $-9=2b+a\ \Rightarrow\ -9=2\times-5+a\Rightarrow a=1$ $\therefore 6-p=2\Rightarrow p=4$	[M1A1
	$q=4\times-\frac{1}{2}+9=7$	M1A1]
(b)	$6x^2-x(4x+9)=5\Rightarrow 6x^2-4x^2-9x-5=0\Rightarrow 2x^2-9x-5=0$	M1
	$2x^2-9x-5=0\Rightarrow(2x+1)(x-5)=0\Rightarrow x=5$	M1A1
	$y=4\times5+9=29\Rightarrow(5,\ 29)$	A1 [4]
		Total 8 marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 3 of 19 · Question 2 part a · 4 marks

2023 JUN · Paper 01 (regional R) · extract 3 of 4 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	Substitutes $\left(-\frac{1}{2}\right)$ into the quadratic to find the value of q (or y)
	A1	For $q = 7$
	M1	For using the linear equation with y = their q and $x = -\frac{1}{2}$ to find the value of p
	A1	For $p = 4$
	ALT	
	M1	Substitutes the linear expression for y and $\left(-\frac{1}{2}\right)$ into the quadratic to find the value of p
	A1	For $p = 4$
	M1	For using the linear equation with their p and $x = -\frac{1}{2}$ to find the value of q
	A1	For $q = 7$
	ALT 2	
	M1A1	For the complete method which must be shown: • Substituting the linear expression for y into the quadratic • Writing $(2x + 1)(ax + b) = 0$ and expanding • Comparing coefficients for form equations in a, b and p • Solving simultaneous equations to find a, b and p
	M1	For using the linear equation with their p and $x = -\frac{1}{2}$ to find the value of q
	A1	For $q = 7$
(b)	M1	For substituting the linear equation (with their value of p) into the quadratic equation to form a 3TQ in x
	M1	For solving their 3TQ, must be a 3TQ. Independent of the first M. See general guidance on what constitutes an attempt to solve a 3TQ.
	A1	For $x = 5$
	A1	For the other solution to the equations. Does not have to be given as coordinates.
	M1M1A1 may be awarded for correct work in (a) if ALT2 was used, then award the final A mark as per scheme.	

Type MS · Where a line meets a curve (simultaneous equations)

Answer 3 of 19 · Question 2 part a · 4 marks

2023 JUN · Paper 01 (regional R) · extract 4 of 4 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 4 of 19 · Question 2 part b · 4 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

2(a)	$6\times\left(-\frac{1}{2}\right)^2-\left(-\frac{1}{2}\right)q=5\Rightarrow\frac{3}{2}+\frac{q}{2}=5\Rightarrow q=7$	M1A1
	$y=px+9\Rightarrow 7=p\times-\frac{1}{2}+9\Rightarrow p=4$	M1A1 [4]
	ALT	
	$6x^2-x(px+9)=5\Rightarrow\frac{3}{2}-\frac{p}{4}+\frac{9}{2}-5=0\Rightarrow p=4$ $y=px+9\ \Rightarrow\ q=4\times-\frac{1}{2}+9\Rightarrow q=7$	[M1A1 M1A1]
	ALT 2	
	$6x^2-x(px+9)-5=0$ and $(2x+1)(ax+b)=0\Rightarrow 2ax^2+x(2b+a)+b=0$ $6-p=2a$ and $b=-5$ $-9=2b+a\ \Rightarrow\ -9=2\times-5+a\Rightarrow a=1$ $\therefore 6-p=2\Rightarrow p=4$ $q=4\times-\frac{1}{2}+9=7$	[M1A1 M1A1]
(b)	$6x^2-x(4x+9)=5\Rightarrow 6x^2-4x^2-9x-5=0\Rightarrow 2x^2-9x-5=0$	M1
	$2x^2-9x-5=0\Rightarrow(2x+1)(x-5)=0\Rightarrow x=5$	M1A1
	$y=4\times5+9=29\Rightarrow(5,\ 29)$	A1 [4]
		Total 8 marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 4 of 19 · Question 2 part b · 4 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks
2(a)	$6\times\left(-\frac{1}{2}\right)^2-\left(-\frac{1}{2}\right)q=5\Rightarrow\frac{3}{2}+\frac{q}{2}=5\Rightarrow q=7$	M1A1
	$y=px+9\Rightarrow 7=p\times-\frac{1}{2}+9\Rightarrow p=4$	M1A1 [4]
	ALT	
	$6x^2-x(px+9)=5\Rightarrow\frac{3}{2}-\frac{p}{4}+\frac{9}{2}-5=0\Rightarrow p=4$	[M1A1
	$y=px+9\ \Rightarrow q=4\times-\frac{1}{2}+9\Rightarrow q=7$	M1A1]
	ALT 2	
	$6x^2-x(px+9)-5=0$ and $(2x+1)(ax+b)=0\Rightarrow 2ax^2+x(2b+a)+b=0$ $6-p=2a$ and $b=-5$ $-9=2b+a\ \Rightarrow\ -9=2\times-5+a\Rightarrow a=1$ $\therefore 6-p=2\Rightarrow p=4$	[M1A1
	$q=4\times-\frac{1}{2}+9=7$	M1A1]
(b)	$6x^2-x(4x+9)=5\Rightarrow 6x^2-4x^2-9x-5=0\Rightarrow 2x^2-9x-5=0$	M1
	$2x^2-9x-5=0\Rightarrow(2x+1)(x-5)=0\Rightarrow x=5$	M1A1
	$y=4\times5+9=29\Rightarrow(5,\ 29)$	A1 [4]
		Total 8 marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 4 of 19 · Question 2 part b · 4 marks

2023 JUN · Paper 01 (regional R) · extract 3 of 4 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	Substitutes $\left(-\frac{1}{2}\right)$ into the quadratic to find the value of q (or y)
	A1	For $q = 7$
	M1	For using the linear equation with y = their q and $x = -\frac{1}{2}$ to find the value of p
	A1	For $p = 4$
	ALT	
	M1	Substitutes the linear expression for y and $\left(-\frac{1}{2}\right)$ into the quadratic to find the value of p
	A1	For $p = 4$
	M1	For using the linear equation with their p and $x = -\frac{1}{2}$ to find the value of q
	A1	For $q = 7$
	ALT 2	
	M1A1	For the complete method which must be shown: • Substituting the linear expression for y into the quadratic • Writing $(2x + 1)(ax + b) = 0$ and expanding • Comparing coefficients for form equations in a, b and p • Solving simultaneous equations to find a, b and p
	M1	For using the linear equation with their p and $x = -\frac{1}{2}$ to find the value of q
	A1	For $q = 7$
(b)	M1	For substituting the linear equation (with their value of p) into the quadratic equation to form a 3TQ in x
	M1	For solving their 3TQ, must be a 3TQ. Independent of the first M. See general guidance on what constitutes an attempt to solve a 3TQ.
	A1	For $x = 5$
	A1	For the other solution to the equations. Does not have to be given as coordinates.
	M1M1A1 may be awarded for correct work in (a) if ALT2 was used, then award the final A mark as per scheme.	

Type MS · Where a line meets a curve (simultaneous equations)

Answer 4 of 19 · Question 2 part b · 4 marks

2023 JUN · Paper 01 (regional R) · extract 4 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 5 of 19 · Question 6 part c · 5 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

Question number		
6 (a)	$f(3) = 27 + 9p + 9 - 30 + q = 0$	M1 A1
	$9p + q + 6 = 0$ *	A1 cso
		(3)
(b)	$f(-p) = -p^3 + p^2(p+1) + 10p + q = 0$	M1 A1
	$p^2 + 10p + q = 0$ *	A1 cso
		(3)
(c)	$p^2 + 10p - 9p - 6 = 0$	M1
	$p^2 + p - 6 = 0$	A1
	$(p+3)(p-2) = 0$	M1
	$p = 2 \quad q = -24$	A1 A1
		(5)
(d)	$(x+a)(x-3)(x+2)$	
	So $-3 \times 2 \times a = -24 \quad a = 4$	M1
	$(x+4)(x-3)(x+2)$	A1
		(2)
		Total 13 marks

Part	Mark	Guidance
General guidance for marking parts (a) and (b) • For the award of full marks in parts (a) and/or (b) you **must** see = 0 used in a line of working **before** the final answer. • If a candidate does not use = 0 in either parts (a) or (b) [except in the final line – which is a given answer] deduct the M mark (and the subsequent A marks) in only the **first** occurrence of the absence.		
(a)	**M1**	For using $f(\pm 3) = 0$ in the given equation set = 0
	A1	For obtaining the correct unsimplified expression: $27 + 9p + 9 - 30 + q = 0$
	A1 cso	For obtaining the given equation $9p + q + 6 = 0$ * **Note: This is a show question. There must be no errors seen.**
(b)	**M1**	For use of $f(\pm p) = 0$ in the given equation set = 0
	A1	For obtaining the correct unsimplified expression: $-p^3 + p^2(p+1) + 10p + q = 0$

Type MS · Where a line meets a curve (simultaneous equations)

Answer 5 of 19 · Question 6 part c · 5 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

	A1 cso	For obtaining the correct given equation $p^2+10p+q=0$ * **Note: This is a show question. There must be no errors seen.**
(c)	**M1**	For attempting to solve the given two equations simultaneously to achieve a 3TQ in either p or q only. E.g. substitutes $q=\mp 9p\mp 6$ or $\left[p=\frac{\mp q\mp 6}{9} \text{ and } p^2=\frac{(\mp q\mp 6)^2}{81}\right]$ into $p^2+10p+q=0$ This mark may be implied by the correct 3TQ
	A1	For the correct 3TQ $p^2+p-6=0$ or $q^2+3q-504=0$
	M1	For an attempt to solve their 3TQ in either p or q using factorisation, use of the formula or completing the square. See general guidance for the definition of an attempt. For example: $(p+3)(p-2)=0\Rightarrow p=\ldots$ or $(q+24)(q-21)=0\Rightarrow q=\ldots,\ \ldots$ If a candidate uses their calculator to solve their 3TQ, the final values must be correct for the award of this mark unless a valid method is seen.
	A1	For either the correct value of p OR the correct value of q $p=2$ or $q=-24$ Condone the presence $p=-3$, and/or $q=21$
	A1	For both the correct value of p (= 2) AND the correct value of q (= –24) Must reject $p=-3$, and/or $q=21$ if seen.
(d)	**M1**	[f(x) = $(x+a)(x-3)(x+'2')$] For attempting to find the value of a $-3\times'2'\times a='-24'\Rightarrow a=\ldots$ OR For an attempt using division with their values of p and q $x^2-x-6\overline{)x^3+3x^2-10x-24}$ or $x-3\overline{)x^3+3x^2-10x-24}$ Allow a quotient of $x+b$ or x^2+6x+b where b is a constant.
	A1	For the correct factorised expression $(x+4)(x-3)(x+2)$ which must be written out in full on one line.

Type MS · Where a line meets a curve (simultaneous equations)

Answer 5 of 19 · Question 6 part c · 5 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 6 of 19 · Question 2 part c · 5 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

	$\frac{\quad}{2}$ is sufficient working
A1	For $19-11\sqrt{3}$ (Allow $a=19,\ b=-11$) As this result can be achieved with a calculator, there can be no incorrect working shown for this mark to be awarded.
ALT	
M1	For $\left(3\sqrt{3}+5\right)\left(a+b\sqrt{3}\right)$
dM1	For attempting to multiply out $\left(3\sqrt{3}+5\right)\left(a+b\sqrt{3}\right)$ to reach an expression that can be used to compare coefficients. There may be up to 2 errors or omissions.
A1	For forming 2 correct simultaneous equations and solving correctly to reach $19-11\sqrt{3}$ (Allow $a=19,\ b=-11$). Two common examples are shown in the scheme. As this result can be achieved with a calculator, there can be no incorrect working shown for this mark to be awarded.

Type MS · Where a line meets a curve (simultaneous equations)

Answer 6 of 19 · Question 2 part c · 5 marks

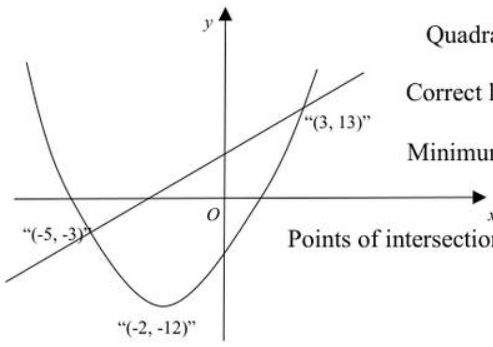
2022 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks
2 a	$(8x+y)(6x+y)\left[=48x^2+14xy+y^2\right]$ oe	B1 (1)
b	$48x^2+14xy+y^2-2(9x^2+15x^2)$ oe $=14xy+y^2$ *	M1 M1 A1 cso* (3)
ALT	eg $3(y\times 3x)+(y\times 5x)+y^2$ or $y(8x+y)+2(3x\times y)$ or $y\times 3x+y\times 5x+y\times(6x+y)$ or $y(6x+y)+y(8x+y)-y^2$ $=14xy+y^2$ *	M1 M1 A1 cso* (3)
c	$(14xy+y^2=1080,\ 48x^2+14xy+y^2=3432)$ $48x^2+1080=3432$ $\left(48x^2=2352\right)$ $x=7$ $y^2+98y-1080(=0)$ $(y-10)(y+108)(=0)$ $y=10$	M1 A1 M1 M1 A1 (5)
ALT	$(14xy+y^2=1080,\ 48x^2+14xy+y^2=3432)$ $x=\dfrac{1080-y^2}{14y}\Rightarrow 48\left(\dfrac{1080-y^2}{14y}\right)^2+\left(\dfrac{1080-y^2}{14y}\right)14y+y^2=3432$ $\Rightarrow y^4-2160y^2+1166400=0$	M1
	$(y^2-1164)(y^2-100)(=0)$ $\left(\Rightarrow y^2=1164 \text{ or } y^2=100 \text{ or } y=\pm 108 \text{ or } y=\pm 10\right)$	M1 (A1 on ePen)
	$y = 10$	A1 (M1 on ePen)
	$14(10)x+100=1080\Rightarrow x=$	M1
	$x=7$	A1
		Total 9 marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 7 of 19 · Question 2 part b · 5 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Question number		
2 (a)	$(x+2)^2-4-8$ $(x+2)^2-12$ $\quad a=2 \quad b=-12$	M1 A1 (2)
(b)	$x^2+4x-8=2x+7$ $x^2+2x-15=0$ $(x-3)(x+5)=0$ or any valid method $x=3, y=13$ $\quad$ $x=-5, y=-3$	M1 dM1 M1 A1 A1 (5)
(c)	 Quadratic drawn Correct line drawn Minimum labelled Points of intersection labelled	B1 B1 B1 ft B1 ft (4) **[11]**

Part	Mark	Additional Guidance
(a)	M1	Use general guidance, allow an expression of the form $\left(x\pm\frac{4}{2}\right)^2\pm q\pm 8 \quad q\neq 0$
	A1	Correct expression as shown, a and b need not be explicitly stated
(b)	M1	Correctly equates the 2 expressions
	dM1	Rearranges to a 3TQ = 0 (allow any 3TQ if intention of rearrangement is clear)
	M1	Uses any valid method to solve – **see general guidance**
	A1	For either pair of values stated
	A1	For all four values, correctly paired or written as coordinates.
	For the final A1 A1, do not allow recovery of y values from part c.	
(c)	B1	Correctly shaped quadratic curve, with a clear minimum point, drawn anywhere on their axis, mark intention.
	B1	Correct line – **must have a positive y intercept, a positive gradient and a negative x intercept**
	B1ft	Correctly labelled coordinates for their minimum, ft their answer from b, **must correctly ft their answer from a, ie minimum point labelled (- *a*, *b*)**
	B1ft	Correctly labelled coordinates for their intersections.
	The coordinates must be clearly indicated and not inferred from a scale on the graph. Ignore any labelling of intersections with axes.	

Type MS · Where a line meets a curve (simultaneous equations)

Answer 7 of 19 · Question 2 part b · 5 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 8 of 19 · Question 10 part c · 4 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

Number		
10 (a)	$6x-x^2=-\left(x^2-6x\right)$ $-\left(x^2-6x\right)=-\left\{(x-3)^2-9\right\}\Rightarrow \mathrm{f}(x)=-(x-3)^2+9$ $D=-1,\ E=-3$ and $F=9$	M1A1A1 [3]
(b)	(i) $\mathrm{f}(x)_{\max}=9$ (ii) $x=3$	B1ft B1ft [2]
(c)	$6x-x^2=x^2-4x+8\Rightarrow 2x^2-10x+8=0$ $2x^2-10x+8=(2x-2)(x-4)\Rightarrow x=1,\ x=4$ $y=5,\ y=8$ Coordinates are (1, 5) and (4, 8)	M1 M1A1 A1 [4]
(d)	$\text{Area}=\int_1^4\left(6x-x^2\right)\mathrm{d}x-\int_1^4\left(x^2-4x+8\right)\mathrm{d}x=\int_1^4\left[-2x^2+10x-8\right]\mathrm{d}x$ $=\left[\frac{-2x^3}{3}+\frac{10x^2}{2}-8x\right]_1^4$ $=\left(\frac{-2\times4^3}{3}+\frac{10\times4^2}{2}-8\times4\right)-\left(\frac{-2\times1^3}{3}+\frac{10\times1^2}{2}-8\times1\right)=9\ (\text{units}^2)$	M1 M1 M1A1 [4]
		Total 13 marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 8 of 19 · Question 10 part c · 4 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 9 of 19 · Question 7 part c · 4 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

7(a)	$x^2-9x+14=\left(x-\frac{9}{2}\right)^2+14-\frac{81}{4}=\left(x-\frac{9}{2}\right)^2-\frac{25}{4}$	M1
	$a=-\frac{9}{2},\ b=-\frac{25}{4}$ oe	A1 (2)
(b)	(i) least value of $f(x)=-\frac{25}{4}$	B1ft
	(ii) least value when $x=\frac{9}{2}$	B1ft (2)
(c)	$x+5=x^2-9x+14$	M1
	$x^2-10x+9=0\Rightarrow(x-9)(x-1)=0$	M1
	Points are $(9,14)$ $(1,6)$	A1A1 (4)
(d)	Area $\int_1^9\left((x+5)-\left(x^2-9x+14\right)\right)dx=\int_1^9\left(-x^2+10x-9\right)dx$	M1
	$=\left[-\frac{x^3}{3}+5x^2-9x\right]_1^9$	M1A1
	$=(-243+405-81)-\left(-\frac{1}{3}+5-9\right)=85\frac{1}{3}$	M1A1 (5)
		[13]

Part	Mark	Notes
(a)	M1	For attempting to complete the square to achieve as a minimum $x^2-9x+14=\left(x\pm\frac{9}{2}\right)^2+14-k$ where k is a constant
	A1	For the correct expression $x^2-9x+14=\left(x-\frac{9}{2}\right)^2-\frac{25}{4}$ or $a=-\frac{9}{2},\ b=-\frac{25}{4}$ oe stated
(b)(i)	B1ft	For the correct value $f(x)=-\frac{25}{4}$ follow through their value of '$-\frac{25}{4}$'
(ii)	B1ft	For the correct value of $x=\frac{9}{2}$ provided they have $\left(x-\frac{9}{2}\right)^2$ …in part (a). Follow through their value of '$\frac{9}{2}$'

Type MS · Where a line meets a curve (simultaneous equations)

Answer 9 of 19 · Question 7 part c · 4 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

(c)	M1	For equating the equation of the line with the equation of C $x+5=x^2-9x+14\Rightarrow x^2-10x+9=0$ **and** attempting to form a 3TQ
	M1	Attempts to solve their 3TQ by any method, provided it is the result of equating the line with C $x^2-10x+9=0\Rightarrow(x-9)(x-1)=0$
	A1	For the correct coordinates of **either** $(9,14)$ or $(1,6)$
	A1	For both correct pairs of coordinates $(9,14)$ and $(1,6)$
(d)	M1	For a correct expression for the required area with both limits correct. (ft their limits from (c)) Award this mark if they have 'curve – line' but otherwise correct. $\int_1^9\left((x+5)-\left(x^2-9x+14\right)\right)\mathrm{d}x=\left[\int_1^9\left(-x^2+10x-9\right)\mathrm{d}x\right]$, accept $\int_1^9\left(x^2-10x+9\right)\mathrm{d}x$ **OR** Area under the trapezium – curve $\frac{1}{2}\times8\times(6+14)-\int_1^9\left(x^2-9x+14\right)\mathrm{d}x$
	M1	For attempting to integrate the equation for the combined expression or the curve only.
	A1	For the correct integrated expression for required area. Ignore limits for this mark – even if they are absent altogether. Area = $=\left[-\frac{x^3}{3}+5x^2-9x\right]_1^9$ accept $\left[\frac{x^3}{3}-5x^2+9x\right]_1^9$ **OR** $\frac{1}{2}\times8\times(6+14)-\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9$ **OR** $\left(\frac{x^2}{2}+5x\right)_1^9-\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9$ or $\left(\frac{x^3}{3}-\frac{9x^2}{2}+14x\right)_1^9-\left(\frac{x^2}{2}+5x\right)_1^9$
	M1	For substituting their limits (x – coordinates from part(c)) into their integrated expression. $=(-243+405-81)-\left(-\frac{1}{3}+5-9\right)=\ldots$ **OR** $80-\left[\left(\frac{9^3}{3}-\frac{9\times9^2}{2}+14\times9\right)-\left(\frac{1^3}{3}-\frac{9\times1^2}{2}+14\times1\right)\right]=\ldots$ **OR** $\left[\left(\frac{9^2}{2}+5\times9\right)-\left(\frac{1^2}{2}+5\times1\right)\right]-\left[\left(\frac{9^3}{3}-\frac{9\times9^2}{2}+14\times9\right)-\left(\frac{1^3}{3}-\frac{9\times1^2}{2}+14\times1\right)\right]=\ldots$
	A1	For the correct area of $85\frac{1}{3}$ or $\frac{256}{3}$ If they get a value of $-85\frac{1}{3}$ they must give a **final** value of $85\frac{1}{3}$ for this mark.

Type MS · Where a line meets a curve (simultaneous equations)

Answer 9 of 19 · Question 7 part c · 4 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question Number	Scheme	Marks
	e^x	

Type MS · Where a line meets a curve (simultaneous equations)

Answer 10 of 19 · Question 5 part b · 2 marks

2023 JAN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

5 a	$\left(y=1+\frac{k}{2}x\Rightarrow\right)$ eg $kx^2-x\left(1+\frac{k}{2}x\right)+(k+1)x-1=0$ or eg $\frac{kx^2+(k+1)x-1}{x}=\frac{k}{2}x+1$	M1
	$kx^2-x-\frac{k}{2}x^2+kx+x-1=0$ or $2kx^2+2(k+1)x-2=kx^2+2x$ oe And an attempt to simplify must be made.	M1
	$kx^2+2kx-2=0$ oe	A1
	$\left(b^2-4ac=0\Rightarrow\right)("2k")^2-4("k")("-2")=0$ or $(x+1)^2=1+\frac{2}{k}$	M1
	$4k(k+2)=0$ or $k+2=0$ or $1+\frac{2}{k}=0$	dM1
	$k=-2$	A1 (6)
ALT **Final 3 marks**	Realises if the line intersects the curve only once, it is a tangent at point A – final 3 marks $y=\frac{kx^2+(k+1)x-1}{x}=\left(kx+k+1-\frac{1}{x}\right)$ and $\left(\frac{dy}{dx}=\right)\frac{k}{2}$	M1
	$\left(\frac{dy}{dx}=\right)k(+0)+\frac{1}{x^2}=\frac{k}{2}\Rightarrow x^2=-\frac{2}{k}$	
	$k\left(-\frac{2}{k}\right)+2k\sqrt{-\frac{2}{k}}-2=0\Rightarrow 2k\sqrt{-\frac{2}{k}}=4\Rightarrow 4k^2\left(-\frac{2}{k}\right)=16\Rightarrow -8k=16$	dM1
	$k=-2$	A1
b	When $k=-2$ $-2x^2-4x-2=0$ or $(x+1)^2=1+\frac{2}{-2}$	
	$(x+1)^2=0$	M1
	$x=-1\Rightarrow y=2$ So $(-1,2)$	A1 (2)
		Total 8 marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 10 of 19 · Question 5 part b · 2 marks

2023 JAN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For correctly substituting $y=1+\frac{k}{2}x$ into $kx^2-xy+(k+1)x=1$
	M1	For a complete expansion of $x\left(1+\frac{k}{2}x\right)$ and $(k+1)x$. Allow one error in the expansion. For multiplying the equation throughout by $2x$ or $-2x$, x or $-x$ to eliminate any algebraic denominators, allow one processing error. There must be an attempt to simplify follow either of these steps. If any other manipulation is seen, full marks can be awarded if the steps lead to a correct equation or a three term quadratic with one processing error made. If there is relevant work, not leading to either of these, please send to review.
	A1	For a correct three term quadratic. oe
	M1	For **clear and correct** use of $b^2-4ac=0$ or a **full and correct** completing the square process, using their expressions for a, b and c. $a,b,c\neq 0$
	dM1	For factorising or any attempt to solve their quadratic in k, or setting their $1+\frac{2}{k}=0$ See general guidance for acceptable attempt to solve a quadratic. For the case of a 2TQ with terms in k and k^2 (which a fully correct solution delivers), we need to see a correct factorisation in the case of this mark or, as the candidate is told k is no zero, they may also correctly divide throughout by k. Dependent on the previous method mark. Allow this mark to be implied if k is correct. Otherwise, the method must be shown to gain this mark.
	A1	$k=-2$, must dismiss $k=0$, if found.
ALT	**States and use the gradient of C is $\frac{k}{2}$.**	
	M1	As main scheme.
	M1	As main scheme.
	A1	As main scheme.
	M1	Differentiates C and puts the gradient = $\frac{k}{2}$ to get an equation of the form $mk+\frac{n}{x^2}=\frac{k}{2}$ $m,n\neq 0$ (doesn't need to be simplified) but must lead to an expression of the form $x^2=-\frac{l}{k}$ $l\neq 0$
	dM1	Substitutes their expression for x^2 into their quadratic equation and rearranges to give an expression of the form $-pk=q$ or $pk=-q$ $p>0,k>0$
	A1	$k=-2$ (must be the only solution given).
(b)	M1	For correct substitution of their value of k into their 3 term quadratic from part a and a minimally acceptable attempt to solve for x. See general guidance.
	A1	For $(-1,2)$ or $x=-1$ and $y=2$

Type MS · Where a line meets a curve (simultaneous equations)

Answer 10 of 19 · Question 5 part b · 2 marks

2023 JAN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks
	$x = 3$	B1 B1

Type MS · Where a line meets a curve (simultaneous equations)

Answer 11 of 19 · Question 10 part a · 4 marks

2023 JAN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

10 a	$(y-1)+y^2=11 \Rightarrow y^2+y-12=0$ or $x^2+\left(x^2+1\right)^2=11 \Rightarrow x^4+3x^2-10=0$	M1
	$(y-3)(y+4)=0$ or $x^2+5 \quad x^2-2=0$	M1
	$(y=3 \quad y=-4) \quad \Rightarrow x^2=2 \quad x^2=-5$	M1
	For stating the x coordinate of A is $-\sqrt{2}$ **and** the x coordinate of B is $\sqrt{2}$	A1 (4)
b	$\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(11-x^2)-(x^2+1)^2\ \mathrm{d}x$ or $\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(x^2+1)^2-(11-x^2)\mathrm{d}x$	M1
	$\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(-x^4-3x^2+10)\ \mathrm{d}x"$	A1ft
	$(\pi)\left[-\frac{x^5}{5}-x^3+10x\right]_{(-\sqrt{2})}^{(\sqrt{2})}$	M1
	$(\pi)\left[\left(-\frac{("\sqrt{2}")^5}{5}-("\sqrt{2}")^3+10"\sqrt{2}"\right)-\left(-\frac{("-\sqrt{2}")^5}{5}-("-\sqrt{2}")^3-10"\sqrt{2}"\right)\right]=63.98$	M1 A1 (5)
ALT		
	$\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(11-x^2)\ \mathrm{d}x-\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(x^2+1)^2\ \mathrm{d}x$ or $\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(x^2+1)^2\ \mathrm{d}x-\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(11-x^2)\ \mathrm{d}x$	{M1}
	$\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(11-x^2)\ \mathrm{d}x-\pi\int_{"-\sqrt{2}"}^{"\sqrt{2}"}(x^4+2x^2+1)\ \mathrm{d}x$	{A1ft}
	$(\pi)\left[11x-\frac{x^3}{3}\right]_{(-\sqrt{2})}^{(\sqrt{2})}-(\pi)\left[\frac{x^5}{5}+\frac{2x^3}{3}+x\right]_{(-\sqrt{2})}^{(\sqrt{2})}$	{M1}
	$(\pi)\left[\left(11("\sqrt{2}")-\frac{("\sqrt{2}")^3}{3}\right)-\left(11("-\sqrt{2}")-\frac{("-\sqrt{2}")^3}{3}\right)\right]$ $-(\pi)\left[\left(\frac{("\sqrt{2}")^5}{5}+\frac{2("\sqrt{2}")^3}{3}+"\sqrt{2}"\right)-\left(\frac{("-\sqrt{2}")^5}{5}+\frac{2("-\sqrt{2}")^3}{3}"-\sqrt{2}"\right)\right]$	{M1} {A1}
	$=63.98$	(5)
		Total 9 marks

Note: the candidate may also set up the integral as

$2\times\pi\int_0^{\sqrt{2}}(11-x^2)-(x^2+1)^2\ \mathrm{d}x$ or $2\times\pi\int_{-\sqrt{2}}^{0}(11-x^2)-(x^2+1)^2\ \mathrm{d}x$ or

$2\times\left(\pi\int_0^{\sqrt{2}}(11-x^2)\ \mathrm{d}x-\pi\int_0^{\sqrt{2}}(x^2+1)^2\ \mathrm{d}x\right)$ *or* $2\times\left(\pi\int_{-\sqrt{2}}^{0}(11-x^2)\ \mathrm{d}x-\pi\int_{-\sqrt{2}}^{0}(x^2+1)^2\ \mathrm{d}x\right)$

If it is clear that the integral(s) being used at any point needs to be multiplied by 2, all marks can be awarded, follow the same principles in the main or ALT schemes if the multiply by 2 is seen anywhere in the solution.

If multiply by 2 is nowhere present or implied by a final answer, the maximum mark is M0 A0 M1 M1 A0

Type MS · Where a line meets a curve (simultaneous equations)

Answer 11 of 19 · Question 10 part a · 4 marks

2023 JAN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For correctly rearranging to get $x^2 = y - 1$ and correctly substituting to get an equation of the form $ay^2 + by + c = 0 \quad a,b,c \neq 0$ **or** For correctly substituting $y = x^2 + 1$, an attempt to expand $(x^2+1)^2$ and attaining an equation of the form $dx^4 + ex^2 + f = 0 \quad d,e,f \neq 0$
	M1	For a complete attempt to solve either $ay^2 + by + c$ or $dx^4 + ex^2 + f$ by any valid method – see general guidance for definition of minimally acceptable attempt.
	M1	For correctly solving their quadratic equation in y or quartic equation in x **and** for obtaining a value(s) for x^2 or x. Award this mark if the candidate does not state or dismisses a solution where x^2 is negative.
	A1	For stating the x coordinate of A is $-\sqrt{2}$ **and** the x coordinate of B is $\sqrt{2}$ Decimal equivalent allowed.
(b)	M1	For $\pi\int_{-\sqrt{2}}^{\sqrt{2}} (11 - x^2) - (x^2+1)^2 \, dx$ or $\pi\int_{-\sqrt{2}}^{\sqrt{2}} (x^2+1)^2 - (11 - x^2) dx$ Allow use of their limits from part a.
	A1ft	For expanding the bracket and simplifying to give $\pi\int_{-\sqrt{2}}^{\sqrt{2}} (-x^4 - 3x^2 + 10) \, dx$ The ft is only for use of their limits from part a.
	M1	For correct integration of their expressions. There must be at least 3 terms overall and there must be a term in x^4 or y^4. It is not necessary for π or limits to be present for this mark.
	M1	For correct substitution of their limits seen into their changed expression. Each of the limits needs to be substituted into the expression correctly at least once. Allow a fully correct final answer to imply this mark. Brackets must be correct but can be recovered later.
	A1	For awrt 63.98 (Calc: 63.9775…) If a negative value is found and changed at the end to a positive value, this final A mark cannot be awarded.
ALT	M1	For $\pi\int_{-\sqrt{2}}^{\sqrt{2}} (11 - x^2) \, dx - \pi\int_{-\sqrt{2}}^{\sqrt{2}} (x^2+1)^2 \, dx$ or $\pi\int_{-\sqrt{2}}^{\sqrt{2}} (x^2+1)^2 \, dx - \pi\int_{-\sqrt{2}}^{\sqrt{2}} (11 - x^2) \, dx$ Allow use of their limits from part a.
	A1ft	For expanding the bracket and simplifying $\pi\int_{-\sqrt{2}}^{\sqrt{2}} (11 - x^2) \, dx - \pi\int_{-\sqrt{2}}^{\sqrt{2}} (x^4 + 2x^2 + 1) \, dx$ The ft is only for use of their limits from part a.
	M1	For correct integration of their expressions. There must be at least 3 terms overall and there must be a term in x^4 or y^4. It is not necessary for π or limits to be present for this mark.
	M1	For correct substitution of their limits seen into their changed expression. Each of the limits needs to be substituted into each expression correctly at least once. Allow a fully correct final answer, with correct integration shown, to imply this mark. Brackets must be correct but can be recovered later.
	A1	For awrt 63.98 (Calc: 63.9775…) If a negative value is found and changed at the end to a positive value, this final A mark cannot be awarded.

Type MS · Where a line meets a curve (simultaneous equations)

Answer 11 of 19 · Question 10 part a · 4 marks

2023 JAN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 12 of 19 · Question 10 part c · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

10 (a)	$f(1) = 32(1^3) - 33(1) + 1 = 0$ *	B1 cso (1)
(b)	$(x-1)(32x^2+32x-1)=0$ A correct method shown to solve a quadratic	M1
	e.g. $\frac{-32 \pm \sqrt{32^2+4\times32}}{64}$	M1
	$(x=1)$ or $\frac{-4+3\sqrt{2}}{8}$ or $\frac{-4-3\sqrt{2}}{8}$ Accept decimals correct to 3 sf e.g. = 0.0303… or = –1.03 …	A1 A1 (4)
(c)	$\sqrt{x} = \frac{1}{8x}$	M1
	$p = \frac{1}{4}$	A1 (2)

	Notes
(a)	
B1 cso	For substitution of 1 into $f(x)$ to obtain the given result
(b)	
M1	For $(x-1)(32x^2+32x-1)=0$
M1	A correct method shown to solve a quadratic. If an algebraic method is not shown then M0A0A0 is awarded.
A1	For $\frac{-4+3\sqrt{2}}{8}$ Allow 0.0303 or better
A1	For $\frac{-4-3\sqrt{2}}{8}$ Allow -1.03 or better
(c)	
M1	For equating the two equations
A1	For $p = \frac{1}{4}$

Type MS · Where a line meets a curve (simultaneous equations)

Answer 12 of 19 · Question 10 part c · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

(d)	$\pi\int_{\frac{1}{4}}^{a} x\,\mathrm{d}x = \pi\left[\frac{x^2}{2}\right]_{\frac{1}{4}}^{a}$	M1
	$= \pi\left(\frac{a^2}{2} - \frac{1}{32}\right)$	A1
	$\pi\int_{\frac{1}{4}}^{a} \frac{1}{64x^2}\,\mathrm{d}x = \pi\left[-\frac{1}{64x}\right]_{\frac{1}{4}}^{a}$	M1
	$= \pi\left(\frac{1}{16} - \frac{1}{64a}\right)$	A1
	$= \pi\left(\frac{a^2}{2} - \frac{1}{32}\right) - \pi\left(\frac{1}{16} - \frac{1}{64a}\right) = \frac{27\pi}{64}$	dM1
	$32a^3 - 33a + 1 = 0$	A1
	So $a = 1$	A1 (7)
	Alternative Method	
	$\pi\int_{\frac{1}{4}}^{a}\left(x - \frac{1}{64x^2}\right)\mathrm{d}x = \frac{27\pi}{64}$	M1
	$\int_{\frac{1}{4}}^{a}\left(64x - \frac{1}{x^2}\right)\mathrm{d}x = 27$	A1
	$\left[32x^2 + \frac{1}{x}\right]_{\frac{1}{4}}^{a} = 27$	M1 A1
	$\left(32a^2 + \frac{1}{a}\right) - (2+4) = 27$	dM1
	$32a^3 - 33a + 1 = 0$	A1
	So $a = 1$	A1 (7) **[14]**

Type MS · Where a line meets a curve (simultaneous equations)

Answer 12 of 19 · Question 10 part c · 2 marks

2020 NOV · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

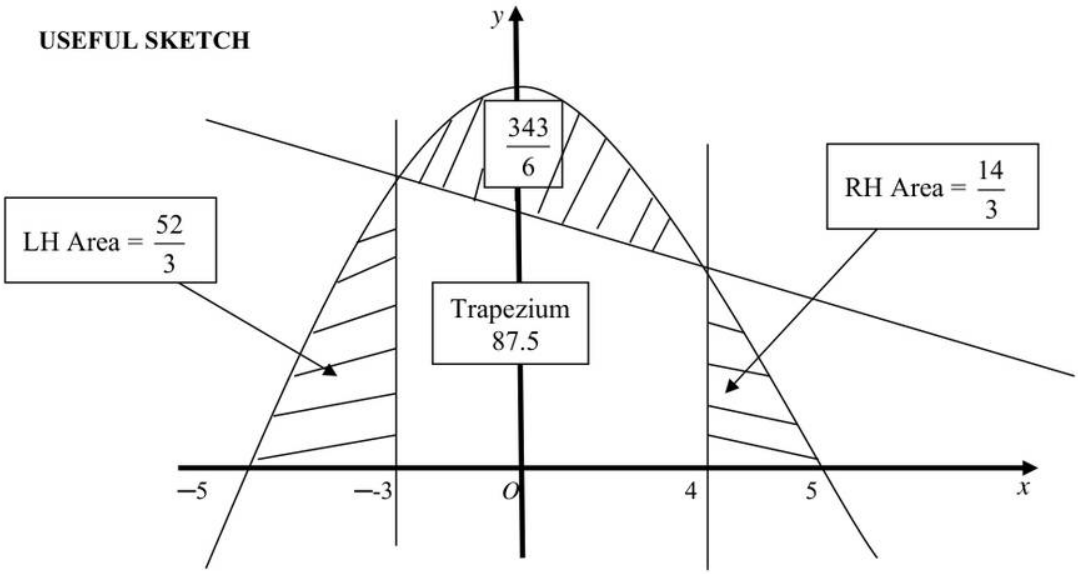
	Notes
(d)	
M1	For an attempt to integrate $\pi\int_{\frac{1}{4}}^{a} x\,\mathrm{d}x$ Ignore limits
A1	For $\pi\left(\frac{a^2}{2}-\frac{1}{32}\right)$
M1	For an attempt to integrate $\pi\int_{\frac{1}{4}}^{a}\frac{1}{64x^2}\,\mathrm{d}x$ Ignore limits
A1	For $=\pi\left(\frac{1}{16}-\frac{1}{64a}\right)$
M1	Dependant on at least one previous M mark being awarded - for subtraction of the two integrals.
A1	For $32a^3-33a+1=0$ oe
A1	For $a=1$
	Alternative
M1	For $\pi\int_{\frac{1}{4}}^{a}\left(x-\frac{1}{64x^2}\right)\mathrm{d}x=\frac{27\pi}{64}$ Ignore limits
A1	For $\int_{\frac{1}{4}}^{a}\left(64x-\frac{1}{x^2}\right)\mathrm{d}x=27$ Ignore limits
M1	For an attempt to integrate $\int_{\frac{1}{4}}^{a}\left(64x-\frac{1}{x^2}\right)\mathrm{d}x$ Ignore limits
A1	$\left[32x^2+\frac{1}{x}\right]_{\frac{1}{4}}^{a}=27$
M1	Dependant on at least one previous M mark being awarded - for correct substitution of the limits
A1	For $32a^3-33a+1=0$ oe
A1	For $a=1$

Type MS · Where a line meets a curve (simultaneous equations)

Answer 13 of 19 · Question 5 part a · 4 marks

2020 JAN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

Number		
5 (a)	$25 - x^2 = 13 - x$ $x^2 - x - 12 = 0$ $(x-4)(x+3) = 0$ $A = (-3, 16)$ $B = (4, 9)$	M1 M1 A1 A1 (4)
(b)	$\int_{-5}^{5}(25-x^2)\mathrm{d}x - \left[\int_{-3}^{4}(25-x^2)\,\mathrm{d}x - \frac{1}{2}(16+9)\times 7\right]$ $\left[25x - \frac{x^3}{3}\right]_{-5}^{5} - \left\{\left[25x - \frac{x^3}{3}\right]_{-3}^{4} - 87.5\right\}$ $\left(\frac{250}{3} + \frac{250}{3}\right) - \left[\left(100 - \frac{64}{3}\right) - (-75+9) - 87.5\right]$ $\frac{219}{2} = (109.5)$	M1 A1 M1 A1 B1 M1 A1 (7)
	Alternative (b) $\int_{-5}^{5}(25-x^2)\mathrm{d}x - \int_{-3}^{4}(12-x^2+x)\,\mathrm{d}x$ $\left[25x - \frac{x^3}{3}\right]_{-5}^{5} - \left[12x - \frac{x^3}{3} + \frac{x^2}{2}\right]_{-3}^{4}$ $\left(\frac{250}{3} + \frac{250}{3}\right) - \left(\frac{104}{3} + \frac{45}{2}\right)$ $\frac{219}{2} = (109.5)$	**[11]** M1 A1 M1 A1 A1 M1 A1 (7)



Type MS · Where a line meets a curve (simultaneous equations)

Answer 13 of 19 · Question 5 part a · 4 marks

2020 JAN · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	For setting the given equation of the curve = given equation of the line $25-x^2=13-x$ and attempting to form a 3TQ $x^2-x-k=0$ (k is an integer) Ignore the absence of = 0 if further work shows that they are attempting to solve a 3TQ = 0
	M1	For attempting to solve their 3TQ See general guidance for the definition of an attempt.
	A1	For **either** $(-3,\ 16)$ or $(4,\ 9)$
	A1	For **both** $(-3,\ 16)$ and $(4,\ 9)$
(b)		There are two ways to calculate this area. In each case; The first M mark is for a correct strategy (allow ft from (a) in their limits) The first A mark (M mark in Epen) is a fully correct strategy with correct limits The second M mark is for an attempt to integrate The second A mark is for a fully correct integration – ignore limits for this mark. The B mark (and A mark in Epen) is for the area of the trapezium of 87.5 seen anywhere. The third M mark is for substituting in their limits The final A mark is the correct answer only.
	Method 1 – Trapezium + two sides	
	M1	For an attempt at the correct strategy to find the area. Allow for this mark a correct statement with using their limits correctly. This may well be seen at the end when they combine individual areas. $(A=)\frac{1}{2}('16'+'9')\times'7'+\int_{'4'}^{'5'}\left(25-x^2\right)\,\mathrm{d}x+\int_{'-5'}^{'-3'}\left(25-x^2\right)\,\mathrm{d}x$ OR $(A=)\int_{-3}^{4}(13-x)\,\mathrm{d}x+\int_{'4'}^{'5'}\left(25-x^2\right)\,\mathrm{d}x+\int_{'-5'}^{'-3'}\left(25-x^2\right)\,\mathrm{d}x$
	A1	Fully correct expression with correct limits. $(A=)\frac{1}{2}(16+9)\times7+\int_{4}^{5}\left(25-x^2\right)\,\mathrm{d}x+\int_{-5}^{-3}\left(25-x^2\right)\,\mathrm{d}x$ OR $(A=)\int_{-3}^{4}(13-x)\,\mathrm{d}x+\int_{'4'}^{'5'}\left(25-x^2\right)\,\mathrm{d}x+\int_{'-5'}^{'-3'}\left(25-x^2\right)\,\mathrm{d}x$
	M1	For an attempt to integrate their expression for area. (Follow General Guidance for the definition of an attempt) Ignore limits for this mark
	A1	For a fully correct integrated expression for the Area with a correct expression for the trapezium (**Ignore limits for this mark**.) $\left[13x-\frac{x^2}{2}\right],\ \left[25x-\frac{x^3}{3}\right],\ \left[25x-\frac{x^3}{3}\right]$ OR $\frac{1}{2}(16+9)\times7,\ \left[25x-\frac{x^3}{3}\right],\ \left[25x-\frac{x^3}{3}\right]$
	B1	For the correct area of the trapezium of 87.5. Award wherever seen. $\frac{343}{6}$ seen implies B1 If not seen explicitly, this can be implied from a correct final answer. **This is an A mark in Epen**

Type MS · Where a line meets a curve (simultaneous equations)

Answer 13 of 19 · Question 5 part a · 4 marks

2020 JAN · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

	M1	For an attempt to substitute **their** limits into **their integrated** expression.
	A1	For the correct final area of $A = \frac{219}{2}$ oe
	Method 2 – Using the area under the whole curve between −5 and 5; minus the area of the curve between −4 and 3; plus the area of the trapezium	
	M1	For an attempt at the correct strategy to find the area Allow for this mark, the correct strategy with **their** limits This may well be seen at the end when they combine individual areas. $(A=)\int_{-5}^{5}(25-x^2)\,dx-\int_{'-3'}^{'4'}(25-x^2)\,dx+\frac{1}{2}('16'+'9')\times'7'$ OR $(A=)\int_{-5}^{5}(25-x^2)\,dx-\int_{'-3'}^{'4'}(25-x^2)\,dx+\int_{'-3'}^{'4'}(13-x)\,dx$ OR $(A=)\int_{-5}^{5}(25-x^2)\,dx-\int_{'-3'}^{'4'}(12+x-x^2)\,dx$
	A1	For the correct expression with correct limits $(A=)\int_{-5}^{5}(25-x^2)\,dx-\int_{'-3'}^{'4'}(25-x^2)\,dx+\frac{1}{2}('16'+'9')\times'7'$ OR $(A=)\int_{-5}^{5}(25-x^2)\,dx-\int_{'-3'}^{'4'}(25-x^2)\,dx+\int_{'-3'}^{'4'}(13-x)\,dx$ OR $(A=)\int_{-5}^{5}(25-x^2)\,dx-\int_{'-3'}^{'4'}(12+x-x^2)\,dx$
	M1	For an attempt to integrate their expression for area. (Follow General Guidance for the definition of an attempt) Ignore limits for this mark
	A1	For a fully correct integrated expression for the Area with a correct expression for the trapezium. Accept this seen as individual parts Ignore limits for this mark. $\left[25x-\frac{x^3}{3}\right],\quad\left[25x-\frac{x^3}{3}\right],\quad\left[13x-\frac{x^2}{2}\right]$ OR $\left[25x-\frac{x^3}{3}\right],\quad\left[25x-\frac{x^3}{3}\right],\quad\frac{1}{2}(16+9)\times7$ OR $\left[25x-\frac{x^3}{3}\right],\quad\left[12x+\frac{x^2}{2}-\frac{x^3}{3}\right]$
	B1	For the correct area of the trapezium of 87.5 $\frac{343}{6}$ seen implies B1 Award wherever seen. If not seen explicitly, this can be implied from a correct final answer. **This is an A mark in Epen**
	M1	For an attempt to substitute **their** limits into **their integrated** expression or individual parts..
	A1	For the correct final area of $A = \frac{219}{2}$ oe

Type MS · Where a line meets a curve (simultaneous equations)

Answer 13 of 19 · Question 5 part a · 4 marks

2020 JAN · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 14 of 19 · Question 11 part c · 3 marks

2023 JUN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks
11(a)	$f(x)=10+6x-x^2=-(x^2-6x)+10 \Rightarrow A=-1$	B1
	$f(x)=-\left[(x-3)^2-9\right]+10=-(x-3)^2+19$	M1
	$B=-3,\ C=19$	A1A1
		[4]
(b)	(i) $x=3$	B1ft
	(ii) f $(x)_{\text{greatest}} = 19$	B1ft
		[2]
(c)	$x^2-x+13=10+6x-x^2 \Rightarrow 2x^2-7x+3=0$	M1
	$\Rightarrow (2x-1)(x-3)=0 \Rightarrow x=\frac{1}{2},\ 3$	M1A1
		[3]
(d)	$A=\int_{\frac{1}{2}}^{3}(10+6x-x^2)\,dx-\int_{\frac{1}{2}}^{3}(x^2-x+13)\,dx=\left[\int_{\frac{1}{2}}^{3}-2x^2+7x-3\,dx\right]$	M1
	$A=\left[-\frac{2x^3}{3}+\frac{7x^2}{2}-3x\right]_{\frac{1}{2}}^{3}$	M1A1
	$A=\left[\left(-\frac{2\times3^3}{3}+\frac{7\times3^2}{2}-3\times3\right)-\left(-\frac{2\times0.5^3}{3}+\frac{7\times0.5^2}{2}-3\times0.5\right)\right]$	M1
	$A=\frac{125}{24}$ oe	A1
		[5]
		Total 14 marks

Part	Mark	Notes
(a)	**B1**	For factorising – 1 and finding $A = -1$
	M1	For an attempt to complete the square – (See general guidance)
	A1	For either $B=-3$ or $C=19$
	A1	For both $B=-3$ and $C=19$ Accept all values embedded.
	NB: Correct values following no working – Award full marks in this part.	

Type MS · Where a line meets a curve (simultaneous equations)

Answer 14 of 19 · Question 11 part c · 3 marks

2023 JUN · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

(b)(i)	**B1ft**	$x = 3$ ft their B You may see differentiation to find a maximum x. Allow a correct value of x here even if it does not follow from their working.
(b)(ii)	**B1ft**	f $(x) = 19$ ft their C You may see differentiation to find a maximum x. Allow a correct value of f(x) here even if it does not follow from their working. Allow $y = 19$.
(c)	**M1**	Sets the equation of curve C equal to the equation of curve S and forms a 3TQ
	M1	Attempts to solve their 3TQ using any correct method to find **two** values of x If their 3TQ is incorrect and no working is seen to solve it, withhold this mark.
	A1	For both $x = \frac{1}{2}$, 3 correct Accept coordinates – ignore y values.
	NB: Both correct values following no working – Award full marks in this part.	
(d)	**M1**	For a correct statement with their values of x in the correct position of the intent to integrate the two expressions and **subtract** the result, or to integrate $2x^2 - 7x + 3$ Ft their limits for this mark, the correct way around, but allow recovery at a later stage. If they deal with the two expressions separately, look for subtraction at the end of their solution.
	M1	For an attempt to integrate their combined expression or **two** separate expressions. They must achieve as a minimum $\pm Px^3 \pm Qx^2 \pm Rx$ for their integration whether they integrate two expressions separately (each) or one combined expression. Ignore limits for this mark.
	A1	For the correct integrated expression(s)– ignore limits for this mark Allow $\frac{2x^3}{3} - \frac{7x^2}{2} + 3x$
	M1	For substituting their limits into their integrated expression and subtracting the results of lower limit from upper limit. **Note carefully:** If they have the correct limits and correct integration allow this mark if the **correct** area is obtained without seeing explicit substitution. $\left(\pm\frac{125}{24},\ \pm 5.208...\right)$ If their limits are incorrect or the integration is incorrect or the final area is incorrect then explicit substitution **must** be seen for the award of this mark.
	A1	For the correct area $A = \frac{125}{24}$ oe

Type MS · Where a line meets a curve (simultaneous equations)

Answer 14 of 19 · Question 11 part c · 3 marks

2023 JUN · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

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Type MS · Where a line meets a curve (simultaneous equations)

Answer 15 of 19 · Question 6 part a · 3 marks

2024 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

6(a)	$2x^2 = \frac{1}{4x} \Rightarrow x^3 = \frac{1}{8} \Rightarrow x = \frac{1}{2}$	M1A1
	$y = 2\times\left(\frac{1}{2}\right)^2 = \frac{1}{2} \Rightarrow \left(\frac{1}{2}, \frac{1}{2}\right)$	B1 [3]
(b)	$y = 2x^2 \Rightarrow x^2 = \frac{y}{2}$, $y = \frac{1}{4x} \Rightarrow x^2 = \frac{y^{-2}}{16}$ or $\left(\frac{1}{4y}\right)^2$ o.e.	B1,B1
	$V = \pi\int_{0.5}^{4}\left(\frac{y}{2}\right)\mathrm{d}y - \pi\int_{0.5}^{4}\left(\frac{y^{-2}}{16}\right)\mathrm{d}y$	M1
	$V = \pi\left[\frac{y^2}{4}\right]_{0.5}^{4} - \pi\left[\frac{y^{-1}}{-16}\right]_{0.5}^{4}$	M1A1
	$V = \frac{\pi}{4}\left(4^2 - \left(\frac{1}{2}\right)^2\right) + \frac{\pi}{16}\left(\frac{1}{4} - \frac{1}{0.5}\right) = \frac{245\pi}{64}$	M1A1
	[The decimal equivalent of the area is 12.0264….]	[7]
		Total 10 marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 15 of 19 · Question 6 part a · 3 marks

2024 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

		Ignore limits and π
	M1	For substituting their limits the correct way around into their integrated expression. You must see this if their integrated expression is incorrect, or their limits are incorrect. If the integration and limits are all correct, the correct volume seen (either in exact or in decimal form i.e. 12.0….) scores this mark. Accept partly processed, for example $\frac{\pi}{4}\left(16-\frac{1}{4}\right)+\frac{\pi}{16}\left(\frac{1}{4}-\frac{1}{0.5}\right)$ as long as you can see **four** calculations/terms as above. Ignore π for this mark.
	A1	For $\frac{245\pi}{64}$
	SC rotates around the x axis. Maximum score is B0B0M1M1A0M1A0	
	B0B0	Not available
	M1	For a correct expression for the volume with the correct calculated limits for x $\left(\sqrt{2}\right.$ and $\left.\frac{1}{16}\right)$ and π Accept the **[correct only]** expressions either way around. You may see π added in at the end. That is fine and please award this mark if that is the case. $V=\pi\int_{\frac{1}{16}}^{\sqrt{2}}\left(\frac{x^{-1}}{4}\right)^2\,\mathrm{d}x-\pi\int_{\frac{1}{16}}^{\sqrt{2}}\left(2x^2\right)^2\,\mathrm{d}x$
	M1	For an attempt to integrate one of their two only expressions. $V=\pi\left[-\frac{x^{-1}}{16}-\frac{4x^5}{5}\right]_{\frac{1}{16}}^{\sqrt{2}}$ Ignore π and limits for this mark
	A0	Not available
	M1	For substituting their values into the integrated expression correctly the correct way around. $V=\pi\left[\left(-\frac{\left(\sqrt{2}\right)^{-1}}{16}-\frac{4\left(\sqrt{2}\right)^5}{5}\right)-\left(-\frac{\left(\frac{1}{16}\right)^{-1}}{16}-\frac{4\left(\frac{1}{16}\right)^5}{5}\right)\right]$
	A0	Not available

Type MS · Where a line meets a curve (simultaneous equations)

Answer 15 of 19 · Question 6 part a · 3 marks

2024 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 16 of 19 · Question 2 part a · 3 marks

2025 NOV · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

2(a)	$3-2x=3x^2+9x-17 \Rightarrow 3x^2+11x-20[=0]$ oe eg $x^2+\frac{11}{3}x-\frac{20}{3}[=0]$ eg $(3x-4)(x+5)[=0] \Rightarrow x=\frac{4}{3}$ oe, -5	M1 M1A1 [3]
(b)	$[3-2x \leqslant 3x^2+9x-17 \Rightarrow 3x^2+11x-20 \leqslant 0]$ $x \leqslant$ "-5", $x \geqslant$ "$\frac{4}{3}$" $x \leqslant -5$, $x \geqslant \frac{4}{3}$ oe	M1 A1 [2]
		Total 5 marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 16 of 19 · Question 2 part a · 3 marks

2025 NOV · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	Sets the line = curve and forms the correct 3TQ The correct 3TQ alone would imply this mark. We will condone a missing = 0
	M1	Solves their 3TQ by any minimally acceptable method (see general guidance) Correct values alone for x will **not** imply this and the previous method mark in this question, as the demand of the question, tells the candidates to 'use algebra' We will condone a missing = 0 Not a fully dependent method mark, but is dependent on correctly setting line = curve.
	A1	For both $x=\frac{4}{3}$ **and** -5 Accept for $\frac{4}{3}$, $1.\dot{3}$ or 1.33.... or 1.3^r or 1.33 (or better) or any equivalent fraction. We need to see as a minimum, **both** the correct 3TQ written down **and** a correct method to solve this equation. As a minimum: Factorising - we must see $(3x-4)(x+5)$ or $(3x-4)$ **and** $(x+5)$ separately or **any** correct factorisation. We will condone, for example, $3x^2+11x-20$ followed by $\left(x-\frac{4}{3}\right)(x+5)$ Using the formula, we need a correct substitution into the formula. This doesn't need to be simplified, but allow one such as $\frac{-11\pm\sqrt{121+240}}{6}$ or $\frac{-11\pm\sqrt{361}}{6}$ as minimum. Completing the square, need to see $3\left(x+\frac{11}{6}\right)^2-\frac{361}{12}$ or $3\left(x+\frac{11}{6}\right)^2=\frac{361}{12}$ or $\left(x+\frac{11}{6}\right)^2=\frac{361}{36}$
(b)	M1	Chooses the outside region, for their values of x which if incorrect, must come clearly from part (a). It is possible to attain M1 A1 if the (ful)l working isn't shown in (a). For a re-start, possibly using a calculator, finding the correct values and choosing the region correctly. possible to be awarded M1 A1 for this approach also. For this mark allow < for $\leqslant$ and > for $\geqslant$ Allow $x\leqslant -5,\quad x\geqslant 1.3$ or $x<-5,\quad x>1.3$ Allow eg $\frac{4}{3}\leqslant x\leqslant -5$
	A1	For the correct region specified correctly. We do not need to see the word 'and'. Accept for $\frac{4}{3}$, $1.\dot{3}$ or 1.33.... or 1.3^r or 1.33 (or better) Accept equivalences eg $(-\infty,-5]$ or $[-\infty,-5]$ and $\left[\frac{4}{3},\infty\right]$ or $\left[\frac{4}{3},\infty\right)$ For this mark do not allow < for $\leqslant$ and > for $\geqslant$ **Do not allow** $x\leqslant -5,\quad x\geqslant 1.3$ or $\frac{4}{3}\leqslant x\leqslant -5$

Type MS · Where a line meets a curve (simultaneous equations)

Answer 16 of 19 · Question 2 part a · 3 marks

2025 NOV · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Where a line meets a curve (simultaneous equations)

Answer 17 of 19 · Question 6 part a · 2 marks

2020 JAN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

6 (a)	When $x = \frac{3}{2}$ $y = 3 - \frac{9}{4} = \frac{3}{4}$ and $2y - \frac{3}{2} = 0$	M1
	So $\left(\frac{3}{2}, \frac{3}{4}\right)$ lies on both the line and the curve	A1cso (2)
(b)	$(\pi)\int_0^{\frac{3}{2}}(2x - x^2)^2 dx$	M1
	$= (\pi)\int_0^{\frac{3}{2}}(4x^2 + x^4 - 4x^3)dx$	A1
	$= (\pi)\left[\frac{4x^3}{3} + \frac{x^5}{5} - x^4\right]_0^{\frac{3}{2}}$	dM1
	$= \frac{153(\pi)}{160}$	A1
	$\frac{153\pi}{160} - \frac{1}{3}\pi\left(\frac{3}{4}\right)^2\left(\frac{3}{2}\right) = \frac{27\pi}{40}$	ddM1A1cao (6)
		[8]
ALT:	$\pi\int_0^{\frac{3}{2}}\left((2x - x^2)^2 - \left(\frac{1}{2}x\right)^2\right)dx$	M1
	$\pi\int_0^{\frac{3}{2}}(\frac{15}{4}x^2 - 4x^3 + x^4)\,dx$	A1
	$\pi\left[\frac{5x^3}{4} - x^4 + \frac{x^5}{5}\right]_0^{\frac{3}{2}}$	dM1A1
	$\pi\left[\frac{135}{32} - \frac{81}{16} + \frac{243}{160}\right] = \frac{27\pi}{40}$	ddM1A1cao

(a)

M1 Attempt to show that $\left(\frac{3}{2}, \frac{3}{4}\right)$ lies on the curve and the line. Any valid method including solving the equations allowed.

A1cso Appropriate conclusion following correct work. Verification, as shown, needs a conclusion.

Solving the equations to obtain $\frac{x}{2} = 2x - x^2$ or $y = 4y - 4y^2$ and hence coordinates of A needs no conclusion. M1 for reaching coords, A1 for correct coords (decimals allowed)

Type MS · Where a line meets a curve (simultaneous equations)

Answer 17 of 19 · Question 6 part a · 2 marks

2020 JAN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

Question Number	Scheme	Marks
(b)	**Algebraic integration must be seen – otherwise no marks.** The first 4 marks can be awarded with or without π provided the work is consistent. The first 3 marks can be awarded if no limits are shown.	
M1	Correct integral, with or without π. Limits may be missing – ignore any shown.	
A1	Square the bracket correctly.	
dM1	Attempt the integration of their integrand. The power of at least one term should increase and no power should decrease. Ignore limits.	
A1	Substitute the correct limits and obtain $\frac{153}{160}$ or $\frac{153\pi}{160}$ (0.95625(pi))	
ddM1	Subtract the volume of the cone from their previous answer. Both terms to include π	
A1cao	Correct final answer (0.675pi)	
ALT:	**See above for general instructions re integration**	
M1	Integral must be the difference of 2 squared terms	
A1	Correct integrand after squaring, need not be simplified	
dM1	Attempt the integration of their integrand. The power of at least one term should increase and no power should decrease.	
A1	Correct result	
ddM1	Substitute their limits	
A1cao	Correct final answer.	

Type MS · Where a line meets a curve (simultaneous equations)

Answer 17 of 19 · Question 6 part a · 2 marks

2020 JAN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

Question Number	Scheme	Marks
	$ar^7 \quad 1152$	

Type MS · Where a line meets a curve (simultaneous equations)

Answer 18 of 19 · Question 4 part a.i · 0 marks

2023 JUN · Paper 02 · find part a.i in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Where a line meets a curve (simultaneous equations)

Answer 19 of 19 · Question 4 part a.ii · 4 marks

2023 JUN · Paper 02 · find part a.ii in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.